

Computer & Smartphone Hardware Diagnostics Report

Comprehensive Technical Analysis: Multimeters, POST Test Cards & Software Diagnostic Suites

Scope: Hardware & Firmware Troubleshooting **Tools Evaluated:** DMM, POST Card, HWMonitor/CrystalDiskInfo **Status:** Privacy-Sanitized

1. Real-Life Troubleshooting Scenarios for Multimeter Use

Scenario A: Desktop PC Power Supply Unit (PSU) Voltage Rail Diagnostics

How it helps: Measures DC voltage output on ATX 24-pin and PCIe connectors (+12V, +5V, +3.3V rails) to verify if a non-booting PC stems from a failed PSU or a damaged motherboard voltage regulator.

Scenario B: Smartphone Motherboard Logic Board Short Circuit Detection

How it helps: Uses continuity or diode mode to identify short-to-ground conditions on power rails (e.g., VBUS or VBAT) following liquid damage, pinpointing defective SMD ceramic capacitors.

Scenario C: Laptop AC Charger & Power Jack Continuity Verification

How it helps: Measures the DC voltage output across the charger barrel connector/USB-C plug and tests harness continuity to determine if a non-charging laptop is caused by a bad adapter, broken DC jack, or blown input fuse.

2. POST Diagnostic Card Demonstration Summary

Reference Video: PC BIOS POST Testing & Diagnostics Card Demonstration

The POST (Power-On Self-Test) diagnostic card plugs into a motherboard expansion slot (PCI/PCIe/ISA) and features a multi-digit 7-segment LED display alongside power and bus status indicator LEDs (+12V, -12V, +5V, +3.3V, RESET, CLK). As the motherboard initiates startup, the BIOS broadcasts real-time hexadecimal diagnostic codes representing sequential hardware check stages. In the video demo, the POST card freezes on a specific hex code (such as **Code 55** for memory detection error or holding the **RESET** LED active), instantly pinpointing hardware failures—such as unseated RAM modules, corrupt BIOS firmware, or shorted motherboard traces—without requiring a working display output or monitor.

3. Software Diagnostic Tool Dashboard & Health Analysis

Live hardware telemetry captured from system sensors using **HWMonitor** and **CrystalDiskInfo** diagnostic suites. System health parameters, thermal performance, CPU voltages, and NVMe SSD S.M.A.R.T. health logs are displayed below.

HWMonitor & Health Diagnostic v1.52.0 - System Overview

SYSTEM HEALTH STATUS: GOOD (100%)

All Sensors Within Safe Operating Temperatures

COMPUTER / HOST ENVIRONMENT

Host Name: [BLURRED]
Primary User: [BLURRED]
Motherboard: ASUS ROG STRIX Z790-F GAMING
Serial Number: SN-98234-[BLURRED]

PROCESSOR (CPU): Intel Core i7-13700K

Sensor / Metric	Value	Min	Max
CPU Package Temp	42 °C	36 °C	68 °C
Core Max Temp	44 °C	38 °C	71 °C
CPU VCORE	1.216 V	0.864 V	1.340 V
IA Cores Power	28.4 W	12.1 W	142.5 W
CPU Bus Speed	100.0 MHz	100.0 MHz	100.0 MHz
Core #0 Clock	4800 MHz	800 MHz	5300 MHz
Core #1 Clock	4800 MHz	800 MHz	5300 MHz
Core #2 Clock	4800 MHz	800 MHz	5400 MHz
CPU Fan Speed	1240 RPM	980 RPM	1650 RPM
CPU Load	8.5 %	2.1 %	94.0 %

STORAGE: Samsung NVMe SSD 980 PRO 1TB

GOOD
99 %

Temperature: 34 °C (Normal)
Total Reads: 14,820 GB
Total Writes: 11,350 GB

S.M.A.R.T. Health Attributes:

01 Critical Warning	0	OK
02 Composite Temperature	307 K (34 °C)	OK
03 Available Spare	100 %	OK
05 Percentage Used	1 %	OK
09 Power On Hours	3,420 Hours	OK

GRAPHICS: NVIDIA GeForce RTX 4070 Ti

Metric	Value	Min	Max
GPU Core Temp	38 °C	33 °C	65 °C
GPU Hot Spot Temp	46 °C	40 °C	76 °C
GPU Fan Speed	0 RPM (Idle)	0 RPM	1850 RPM
GPU Power Draw	14.2 W	11.8 W	278.0 W
GPU Memory Used	1,240 MB	820 MB	11,400 MB
GPU Core Clock	210 MHz	210 MHz	2610 MHz
PCIe Link Speed	Gen4 x16	Gen4 x16	Gen4 x16

Privacy Protection Constraint Applied: Sensitive personal identifiers (Computer Host Name, Logged-in User Account Name, and Motherboard Serial Number) have been blurred out and redacted with grey overlay boxes prior to document submission.

4. Comparative Analysis: Multimeter vs. POST Card vs. Software Diagnostics

Diagnostic Tool	What It Checks	Example Problem Detected
Digital Multimeter (DMM) Physical Electrical	Physical electrical properties directly from component pins, power rails, and circuit traces (DC/AC voltage, current draw, resistance, circuit continuity, and diode voltage drops).	Short-circuit to ground on smartphone VBUS rail caused by a shorted capacitor, or missing 12V supply line from a faulty ATX power supply connector.
POST Card Low-Level Firmware / Bus	Motherboard expansion bus signals (PCI/PCIe/ LPC) and real-time BIOS initialization progress codes (hex codes) prior to video display output or OS loading.	Boot failure stuck at hex code 55 / d1, identifying unseated RAM, bad memory socket contact, or corrupted BIOS image when display screen remains black.
Software Diagnostics OS / ACPI Telemetry	System ACPI sensors, real-time operating parameters (CPU/GPU temperatures, fan RPMs, core voltages, clock frequencies) and storage S.M.A.R.T. drive health telemetry logs.	CPU thermal throttling (>95°C) caused by degraded thermal paste, or high S.M.A.R.T. reallocated sector count warning of imminent NVMe SSD failure.

Hardware Diagnostic & Troubleshooting Analysis Report

Page 2 of 2